

Conditional probability practice questions

2-57. Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		shock resistance	
		high	low
scratch resistance	high	70	9
	low	16	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. Determine the following probabilities:

- (a) $P(A)$ (b) $P(B)$
(c) $P(A|B)$ (d) $P(B|A)$

2-58. Samples of a cast aluminum part are classified on the basis of surface finish (in microinches) and length measurements. The results of 100 parts are summarized as follows:

		length	
		excellent	good
surface finish	excellent	80	2
	good	10	8

Let A denote the event that a sample has excellent surface finish, and let B denote the event that a sample has excellent length. Determine:

- (a) $P(A)$ (b) $P(B)$
(c) $P(A|B)$ (d) $P(B|A)$
(e) If the selected part has excellent surface finish, what is the probability that the length is excellent?
(f) If the selected part has good length, what is the probability that the surface finish is excellent?

2-59. The analysis of shafts for a compressor is summarized by conformance to specifications:

		roundness conforms	
		yes	no
surface finish	yes	345	5
conforms	no	12	8

- If we know that a shaft conforms to roundness requirements, what is the probability that it conforms to surface finish requirements?
- If we know that a shaft does not conform to roundness requirements, what is the probability that it conforms to surface finish requirements?

2-60. The following table summarizes the analysis of samples of galvanized steel for coating weight and surface roughness:

		coating weight	
		high	low
surface	high	12	16
roughness	low	88	34

- If the coating weight of a sample is high, what is the probability that the surface roughness is high?
- If the surface roughness of a sample is high, what is the probability that the coating weight is high?
- If the surface roughness of a sample is low, what is the probability that the coating weight is low?

2-61. Consider the data on wafer contamination and location in the sputtering tool shown in Table 2-2. Assume that one wafer is selected at random from this set. Let A denote the event that a wafer contains four or more particles, and let B denote the event that a wafer is from the center of the sputtering tool. Determine:

- $P(A)$
- $P(A|B)$
- $P(B)$
- $P(B|A)$
- $P(A \cap B)$
- $P(A \cup B)$

2-62. A lot of 100 semiconductor chips contains 20 that are defective. Two are selected randomly, without replacement, from the lot.

- (a) What is the probability that the first one selected is defective?
- (b) What is the probability that the second one selected is defective given that the first one was defective?
- (c) What is the probability that both are defective?